

Lecture Note 8: Testing Concepts

14.380: Statistical Methods in Economics, Fall 2026

1 Hypotheses

Hypotheses are statements about the population distribution, which are either true or untrue for the given population.

Example For example, let $X_1, \dots, X_n$ be a random sample from distribution $N(\mu, \sigma^2)$ with σ^2 known and $\mu \in \mathcal{M}$. Suppose our hypothesis is that $\mu \in \mathcal{M}_1$ for some $\mathcal{M}_1 \subset \mathcal{M}$, i.e. $\mathcal{M}_1$ is some subset of $\mathcal{M}$. It is called the *null hypothesis*. It is denoted as $H_0 : \mu \in \mathcal{M}_1$. Then the *alternative hypothesis* is that $\mu \notin \mathcal{M}_1$, i.e. $\mu \in \mathcal{M} \setminus \mathcal{M}_1$. It is denoted as $H_a : \mu \in \mathcal{M} \setminus \mathcal{M}_1$. For example, if $\mathcal{M}_1 = \{\mu : \mu \leq \mu_0\}$ and $\mathcal{M} = \mathbb{R}$, then $H_0 : \mu \leq \mu_0$ and $H_a : \mu > \mu_0$. Or, as another example, if $\mathcal{M}_1 = \{\mu_0\}$, then $H_0 : \mu = \mu_0$ and $H_a : \mu \neq \mu_0$.

If a hypothesis uniquely identifies the distribution of the data, it is called simple. Otherwise, the hypothesis is called composite. In the second example above, the null hypothesis is simple, while the alternative is composite. It is customary to mention both the null and the alternative hypotheses since the full parameter space $\mathcal{M}$ is often unspecified.

2 Testing

We observe a sample from a population and, based on this sample, create a test. Our test is intended to decide whether we accept the null hypothesis or reject it in favor of the alternative. Some people argue that instead of word “accept” it is more appropriate to say “do not reject”. We are not going to emphasize this difference here.

2.1 Critical region

Let X denote our data. Then any test consists of the critical region C , which is a function of our null and alternative hypotheses, such that we accept the null hypothesis if $X \in C$ and reject it if $X \notin C$. For example, if our data is $X = (X_1, \dots, X_n)$, then the critical region might be $C = \{\sum_{i=1}^n X_i < \delta\}$ for some $\delta \in \mathbb{R}$. The value δ in this example might depend both on the null and on the alternative.

In testing, four situations are possible. If H_0 is true and we accept it, then it is a correct decision. If H_0 is true but we reject it, then it is a type 1 error. If H_0 is false but we accept it, then it is a type 2 error. If H_0 is false and we reject it, then it is a correct decision again. So, in addition to correct decisions, there are errors of two types.

2.2 Size and power trade-off

The probability of a type-1 error is called the *size* of the test.

Example (cont.) In the example above, suppose our null hypothesis is $H_0 : \mu = \mu_0$ and our alternative is $H_a : \mu > \mu_0$. Then the natural test is to accept the null hypothesis if the data belongs

to the critical region $C = \{\sum_{i=1}^n X_i < \delta\}$. Then

$$P_{\mu_0}\left\{\sum_{i=1}^n X_i \geq \delta\right\} = P_{\mu_0}\left\{\sqrt{n}(\bar{X}_n - \mu_0)/\sigma \geq \sqrt{n}(\delta/n - \mu_0)/\sigma\right\} = 1 - \Phi(\sqrt{n}(\delta/n - \mu_0)/\sigma),$$

which is a decreasing function of δ . If δ is large, then size of the test is small, which is good. Please note that the size is calculated at the null value (often called “under the null”).

What is the probability of a type-2 error? If true parameter value $\mu > \mu_0$, then

$$P_{\mu}\left(\sum_{i=1}^n X_i < \delta\right) = \Phi(\sqrt{n}(\delta/n - \mu)/\sigma).$$

First, notice that it is a function of true μ . Second, if δ is large, then the probability of a type-2 error is large as well, which is bad.

Thus, there is a trade-off between the probability of a type-1 error and the probability of a type-2 error. This trade-off exists in most practically relevant situations. Before we consider how one should choose the test in light of this trade-off, the introduction of some additional concepts is necessary.

The *power* of the test is defined as the probability of correctly rejecting the null hypothesis. Thus, the power of the test is defined as 1 minus the probability of a type-2 error. Clearly, the power of the test depends on the true parameter value. So, power is usually considered to be a function of the true parameter value on the set of alternatives.

The size of the test also depends on the true parameter value when the null hypothesis is composite. But, instead of considering the size of the test as a function of the true parameter value, the concept of the level of the test is used. We say that the test has *level* α if for any true parameter value in the null hypothesis, the size is not greater than α . The level of the test is defined as the maximum of the size over all possible true parameter values in the null hypothesis. In the example above, the level of the test is $\sup_{\mu \in \mathcal{M}_1} \text{size}(\mu)$.

Once we have some notion of the power of the test and its level, let us consider how to choose the test. Common practice is to fix the level of the test (usually, it is 1, 5, or 10%) and then to choose a test with as much power as possible among all tests of a given level. In this sense the null and the alternative are not treated equally.

Example (cont.) Let us return to our example with a gaussian sample with known variance, where $H_0 : \mu = \mu_0$ and $H_a : \mu > \mu_0$. Suppose we want a test with level 5%. We wish to reject the null when $\sum_i X_i$ is large (that is, the critical region is of the form $C = \{\sum_{i=1}^n X_i < \delta\}$). We equivalently may construct statistics as follows:

$$Z = Z(X, \mu_0) = \frac{\sqrt{n}}{\sigma} (\bar{X} - \mu_0) \sim N(0, 1) \text{ under the null.}$$

This is often called a Z -statistic. Let $Z_{0.95}$ denote the 95%-quantile of a standard normal distribution. Then $\sqrt{n}(\delta/n - \mu_0)/\sigma = Z_{0.95}$ or, our test will accept the null if $Z < Z_{0.95}$. This is the test with exact (finite-sample) size 5%.

Since the power of the test depends on the true parameter value, it is possible that one test has maximal power among all tests with a given level at one parameter value, while another test has maximal power at some other parameter value. So it is possible that there is no *uniformly most powerful test*. In this situation the researcher should consider some additional criteria to choose a test. This observation explains a wide variety of tests suggested in the statistics and econometrics literature. However, we should note that there is an important class of problems where uniformly most powerful tests do exist. We will discuss it next time.

2.3 The Power Function, Power Analysis, and Minimum Detectable Effects

So far we have treated power as a number attached to a single alternative. It is more useful to treat it as a function.

Definition 1. The **power function** of a test with critical region C is

$$\beta(\theta) = P_{\theta}(\text{the test rejects } H_0),$$

viewed as a function of θ over the whole parameter space.

The power function contains everything there is to know about a test: its size is the value (or supremum) of $\beta(\theta)$ over the null, and its power against any alternative is β evaluated there.

Consider the one-sample problem: $X_1, \dots, X_n$ i.i.d. with mean μ and known variance σ^2 , testing $H_0: \mu = \mu_0$ against a two-sided alternative with the test that rejects when $|\sqrt{n}(\bar{X}_n - \mu_0)/\sigma| > z_{1-\alpha/2}$, where z_q denotes the q -th quantile of the standard normal distribution. Suppose the true mean is $\mu = \mu_0 + \tau$. Then $\sqrt{n}(\bar{X}_n - \mu_0)/\sigma$ is (approximately) $N(\sqrt{n}\tau/\sigma, 1)$, and a direct calculation gives

$$\beta(\tau) = \Phi\left(-z_{1-\alpha/2} + \frac{\sqrt{n}\tau}{\sigma}\right) + \Phi\left(-z_{1-\alpha/2} - \frac{\sqrt{n}\tau}{\sigma}\right) \approx \Phi\left(-z_{1-\alpha/2} + \frac{\sqrt{n}|\tau|}{\sigma}\right),$$

where the approximation drops the far-tail term, which is negligible unless τ is very close to zero. The power function equals α at $\tau = 0$ and increases toward one as $|\tau|$ or n grows: larger effects and larger samples are easier to detect.

This calculation can be inverted in two ways, and both inversions are used constantly in empirical practice.

Power analysis (solve for n). Suppose the researcher is designing a study and wants power at least $1 - \kappa$ against an effect of scientific interest τ . Setting $\beta(\tau) \geq 1 - \kappa$ and solving,

$$\frac{\sqrt{n}|\tau|}{\sigma} \geq z_{1-\alpha/2} + z_{1-\kappa} \iff n \geq \frac{(z_{1-\alpha/2} + z_{1-\kappa})^2 \sigma^2}{\tau^2}.$$

With the conventional choices $\alpha = 0.05$ and $1 - \kappa = 0.80$, the constant is $(1.96 + 0.84)^2 \approx 7.85$: the required sample size is about $7.85 \sigma^2 / \tau^2$. Halving the effect one hopes to detect quadruples the required sample.

Minimum detectable effects (solve for τ). Often the design is fixed — the researcher has n observations and cannot get more — and the honest question runs the other way: what is the smallest effect this study can reliably detect? Solving the same inequality for $|\tau|$ yields the *minimum detectable effect*

$$MDE = (z_{1-\alpha/2} + z_{1-\kappa}) \times \frac{\sigma}{\sqrt{n}} \approx 2.8 \times (\text{standard error}),$$

again at the conventional $\alpha = 0.05$ and 80% power. The same logic applies to a two-group experiment: with n units split equally between treatment and control and outcome variance σ^2 in each group, the standard error of the difference in means is $2\sigma/\sqrt{n}$, so the minimum detectable treatment effect is approximately $5.6 \sigma / \sqrt{n}$.

Two lessons are worth internalizing. First, an effect smaller than the MDE is not evidence of no effect; it is an effect the study was never capable of seeing, and “insignificant” results from underpowered studies are close to uninformative. Second, when an underpowered study *does* reject, the estimated effect must be large — larger, typically, than the truth — so statistically significant estimates from small studies systematically exaggerate. Power calculations of this kind are a required component of the pre-analysis plans and grant applications that accompany modern randomized experiments; Duflo, Glennerster, and Kremer (2007) provide a practitioner’s guide.

2.4 P-value

The result of any test is either acceptance or rejection of the null hypothesis. At the same time, it would be interesting to know to what extent we are sure about the result of the test. The concept of the p-value gives us such a measure. The *p-value* is the probability (calculated under the null) of obtaining a sample at least as adverse to the null hypothesis as given. Notice that the p-value is a random variable.

Example (cont.) Let $z = Z(x, \mu_0)$ be the value of the Z -statistic, as defined above, that we see in our data set

$$\text{p-value} = P\{N(0, 1) > z\} = 1 - \Phi(z).$$

Note again, that it is a function of z , and thus is a random variable. By construction, our test rejects the null if the p-value is smaller than 0.05.

If the p-value is much smaller than 0.05, then we can be quite sure that the null hypothesis does not hold. If the p-value is close to 0.05, then we are not so sure. Moreover, reporting the p-value has the advantage that, once the p-value is reported, any researcher can decide for himself whether he or she accepts or rejects the null hypothesis depending on his/her own favorite level of the test.

Let us now emphasize some frequent misunderstandings about the concept of the p-value.

First, a p-value is not the probability that the null is true. There is no such probability at all since parameters are not random according to the frequentist (classical) approach. Second, the p-value is not the probability of falsely rejecting the null. This probability is measured by the size of the test. Third, one minus p-value is not the probability of the alternative being true. Again, there is no such probability since parameters are not random. Finally, the level of the test is not determined by a p-value. Instead, once we know the p-value of the test, the level of the test determines whether we accept or reject the null hypothesis.

Example. Let $X_1, \dots, X_n$ be a random sample from an $N(\mu, \sigma^2)$ distribution. The null hypothesis, H_0 , is that $\sigma^2 = \sigma_0^2$. The alternative hypothesis, H_a , is that $\sigma^2 < \sigma_0^2$. Note that both hypotheses are composite since they contain all possible values of μ . Let us construct a test based on sample variance s^2 . We know that $(n-1)s^2/\sigma^2 \sim \chi^2(n-1)$. Since small values of $(n-1)s^2/\sigma_0^2$ are a sign in favor of the alternative, our critical region should take the form $C = \{(n-1)s^2/\sigma_0^2 > k\}$. Under H_0 , $(n-1)s^2/\sigma_0^2 \sim \chi^2(n-1)$. Then a test with level, say, 5%, accepts the null hypothesis if $(n-1)s^2/\sigma_0^2 > \chi_{0.05}^2(n-1)$, where $\chi_{0.05}^2(n-1)$ denotes the 5%-quantile of $\chi^2(n-1)$. What is the power of this test? Let $\sigma^2 < \sigma_0^2$. Then

$$P_{\sigma^2}\{(n-1)s^2/\sigma_0^2 \leq \chi_{0.05}^2\} = P_{\sigma^2}\{(n-1)s^2/\sigma^2 \leq (\sigma_0^2/\sigma^2)\chi_{0.05}^2\} = F_{\chi^2(n-1)}((\sigma_0^2/\sigma^2)\chi_{0.05}^2),$$

where $F_{\chi^2(n-1)}$ denotes the cdf of $\chi^2(n-1)$. So the power of the test increases as σ^2 decreases. Suppose $n = 101$, $\sigma_0^2 = 1$, and we observe $s^2 = 0.9$. What is the p-value of our test? Let $A \sim \chi^2(n-1)$. Then the p-value equals

$$P\{A \leq (n-1)s^2/\sigma_0^2\} = F_{\chi^2(n-1)}((n-1)s^2/\sigma_0^2) = F_{\chi^2(100)}(100 \cdot 0.9/1) = F_{\chi^2(100)}(90) \approx 0.25.$$

Thus, the test with level 5% does not reject the null hypothesis.

Remark 1 (Multiple testing). So far we have considered testing a single hypothesis. In practice, researchers often test many hypotheses at once — for example, the effect of a treatment on each of m different outcomes. Suppose each of the m null hypotheses is tested at level α . Even if every null hypothesis is true, the probability of making at least one false rejection can be much larger than α : by the union bound (the same logic we used to combine tail probabilities in the concentration inequalities of Lecture Note 3), $P\{\text{at least one false rejection}\} \leq \sum_{j=1}^m P\{\text{reject } H_{0,j}\} \leq m\alpha$, and this upper bound may be nearly attained when the tests are approximately independent. The probability of at least one false rejection is called the *familywise error rate*. The *Bonferroni correction* controls it in a simple way: test each hypothesis at level α/m or, equivalently, reject $H_{0,j}$ only if its p-value is below α/m . Then the familywise error rate is at most $m \cdot (\alpha/m) = \alpha$. The price of this protection is power: each individual test becomes more conservative as the number of hypotheses m grows.

3 Pivotal Statistics

By definition, a statistic is called *pivotal* if its distribution is independent of unknown parameters. Pivotal statistics are useful in testing because one can calculate quantiles of their distributions and, thus, critical values for tests based on these statistics. For example, $(n-1)s^2/\sigma_0^2$ from the example above is pivotal under the null since its distribution does not depend on μ .

3.1 Asymptotic tests

Example. Test of the mean. Let $X = (X_1, \dots, X_n)$ be a random sample from some unknown distribution with finite second moment. The null hypothesis is $H_0 : \mathbb{E}[X_i] = \mu_0$. The alternative is $H_a : \mathbb{E}[X_i] \neq \mu_0$. Both hypotheses are composite. Let us construct a test based on $|\bar{X}_n - \mu_0|$. Large values of $|\bar{X}_n - \mu_0|$ are a sign in favor of the alternative. Thus, our critical region should take the form $C = \{|\bar{X}_n - \mu_0| \leq \delta\}$ for some $\delta > 0$. It is impossible to find the exact distribution of the test statistic in this case but possible to find an *asymptotically pivotal* test statistic, that is, a test statistic with an asymptotic distribution that does not depend on any unknown parameter.

Under the null, $\sqrt{n}(\bar{X}_n - \mu_0) \xrightarrow{d} N(0, \text{Var}(X_i))$. We also have a consistent estimate for $\text{Var}(X_i)$ which is s^2 . Let us define

$$t = (\bar{X}_n - \mu_0) / \sqrt{s^2/n} \xrightarrow{d} N(0, 1) \text{ if the null is true.}$$

This is the well-known t -statistic. We reject the null at 5% level if $|t| > 1.96$ and the p-value is $2\Phi(-|t|)$. This test has *asymptotic size* 5%. That is,

$$P_{H_0}\{\text{reject}\} \rightarrow 0.05 \text{ as } n \rightarrow \infty.$$

Example As another example, let $X_1, \dots, X_m$ and $Y_1, \dots, Y_n$ be independent random samples from two different distributions. We want to test the null hypothesis, H_0 , that $\mathbb{E}[X_i] = \mathbb{E}[Y_i]$ against the alternative, H_a , that $\mathbb{E}[X_i] > \mathbb{E}[Y_i]$. A natural place to start is to note that if the null hypothesis is true, then $\bar{X}_m$ should be close to $\bar{Y}_n$ with high probability. But, in large samples, $\bar{X}_m - \bar{Y}_n$ is approximately distributed as $N(0, \sigma_x^2/m + \sigma_y^2/n)$ with σ_x^2 and σ_y^2 unknown variances of the two distributions. So consider

$$t = \frac{\bar{X}_m - \bar{Y}_n}{\sqrt{s_x^2/m + s_y^2/n}},$$

where s_x^2 and s_y^2 are sample variances. The exact distribution of the t -statistic here is not pleasant. Instead, let us use asymptotic theory. As an exercise, prove that if both n and m increase to infinity we have $t \xrightarrow{d} N(0, 1)$. So we can use the quantiles of a standard normal distribution to form a test with size approximately equal to that of the required level of the test. This gives us a test of “asymptotically the correct size”.

3.2 Bootstrap

Example. Test of the variances. Let $X = (X_1, \dots, X_n)$ be a random sample from some unknown distribution with finite fourth moment. The null hypothesis is $H_0 : \text{Var}(X_i) = \sigma_0^2$. The alternative is $H_a : \text{Var}(X_i) \neq \sigma_0^2$. Both hypotheses are composite. We obviously want to use a statistic based on the sample variance s^2 , but without a normality assumption the exact distribution of s^2 is unavailable. We know that s^2 is consistent and asymptotically gaussian (think about why), but we may be lazy to figure out its asymptotic variance. Imagine we want to use the statistic $z = \sqrt{n}(s^2 - \sigma_0^2)$ and reject when it is too large or too small. We may do the following:

- For $b = 1, \dots, B$ repeat:
 - Draw an i.i.d. sample $X_b^* = (X_{1b}^*, \dots, X_{nb}^*)$ from the set of initial observations $\{X_1, \dots, X_n\}$ with replacement;
 - Calculate s_b^2 , the sample variance of X_b^* ;
 - Calculate $z_b = \sqrt{n}(s_b^2 - s^2)$;
- Order the z_b in ascending order: $z_{(1)} \leq \dots \leq z_{(B)}$;
- For a test of size α , if $z_{(\lfloor \frac{\alpha}{2} B \rfloor)} < z < z_{(\lfloor (1 - \frac{\alpha}{2}) B \rfloor)}$ accept the null, otherwise reject.

Think about why this procedure would give an asymptotically level- α test. We know that under the null $z = \sqrt{n}(s^2 - \sigma_0^2) \xrightarrow{d} N(0, V)$, where the asymptotic variance V can be calculated using the CLT and the delta method. It depends on the first four moments of the random variable X_i (which are unknown).

For the bootstrapped sample, the variance equals the sample variance s^2 , and the first four moments are consistent estimators of the population's first four moments. We have $z^* = \sqrt{n}((s^*)^2 - s^2) \xrightarrow{d} N(0, V^*)$ and $V^* \xrightarrow{p} V$. So the bootstrapped distribution of the z -statistic will asymptotically be the same as the true asymptotic distribution of the z -statistic.

Example. Test of a nonlinear hypothesis. The same logic extends well beyond the variance example. Let $X = (X_1, \dots, X_n)$ be a random sample from some unknown distribution with finite fourth moment. Denote $\mu = \mathbb{E}[X_i]$ and $\sigma^2 = \text{Var}(X_i)$, and let $h(\mu, \sigma^2)$ be a function that is twice continuously differentiable in both arguments with a nonzero derivative at the true parameter values. Suppose we wish to test the hypothesis $H_0 : h(\mu, \sigma^2) = 0$ against $H_a : h(\mu, \sigma^2) \neq 0$.

One way to approach this problem is to define a new parameter $\gamma = h(\mu, \sigma^2)$, for which we have a consistent estimate $\hat{\gamma} = h(\bar{X}_n, s^2)$. We can even prove asymptotic gaussianity via the delta method:

$$\sqrt{n}(\hat{\gamma} - \gamma) \xrightarrow{d} N(0, (\nabla h)' V \nabla h),$$

where $\sqrt{n}(\bar{X}_n - \mu, s^2 - \sigma^2)' \xrightarrow{d} N(0, V)$ and $\nabla h = \left(\frac{\partial h(\mu, \sigma^2)}{\partial \mu}, \frac{\partial h(\mu, \sigma^2)}{\partial \sigma^2} \right)'$. So, in principle, one may obtain a natural estimate of the asymptotic variance of $\hat{\gamma}$ and construct a regular t -statistic to conduct a test. Alternatively, one can employ the nonparametric bootstrap exactly as in the

variance example, approximating the unknown finite-sample distribution of the statistic $z = \sqrt{n}(\hat{\gamma} - 0)$ with that of

$$z^* = \sqrt{n}(\hat{\gamma}^* - \hat{\gamma}), \quad \hat{\gamma}^* = h(\bar{X}_n^*, (s^*)^2).$$

Let us denote all parameters and statistics related to the bootstrapped distribution with stars; for example, $\mu^* = \int x d\hat{F}_n = \mathbb{E}^*[X_i^*] = \bar{X}_n$, where $\hat{F}_n$ is the empirical distribution function of the sample (Lecture Note 4). Several comments are in order.

First, notice the “re-centering” of the null hypothesis in the definition of z^* — and, for that matter, in the variance example above, where we calculated $z_b = \sqrt{n}(s_b^2 - s^2)$ rather than $\sqrt{n}(s_b^2 - \sigma_0^2)$. We approximate the unknown distribution of X_i with the empirical distribution function $\hat{F}_n$. Size is defined under the null, so we assume that the unknown distribution of X_i is such that $H_0 : h(\mu, \sigma^2) = 0$ holds. However, we know for sure that this null does not hold for the empirical distribution $\hat{F}_n$: its mean is $\mu^* = \bar{X}_n$ and its variance is $\sigma^{*2} = \frac{1}{n} \sum_i (X_i - \bar{X}_n)^2$ (almost s^2 ; the difference is of order $O_p(1/n)$ and will be ignored here as too small to care about). From this perspective, the true value of the parameter γ for the empirical distribution is (almost) $\hat{\gamma}$. So, for the bootstrapped samples, we should be testing the null hypothesis that is actually true for $\hat{F}_n$, namely $H_0 : h(\mu^*, \sigma^{*2}) = \hat{\gamma}$.

Second, the justification of bootstrap validity here has the same structure as in the variance example: it is based on the validity of the asymptotic approximation, both for the statistic itself and for its bootstrapped version. Under the bootstrapped distribution,

$$\sqrt{n}(\hat{\gamma}^* - \hat{\gamma}) \xrightarrow{d} N(0, (\nabla h^*)' V^* \nabla h^*),$$

where $\sqrt{n}(\bar{X}_n^* - \mu^*, (s^*)^2 - \sigma^{*2})' \xrightarrow{d} N(0, V^*)$ and $\nabla h^* = \left(\frac{\partial h(\mu^*, \sigma^{*2})}{\partial \mu}, \frac{\partial h(\mu^*, \sigma^{*2})}{\partial \sigma^2} \right)'$. Noticing that $V^* = V + o_p(1)$ and $\nabla h^* = \nabla h + o_p(1)$, the two limit distributions coincide.

Third, this pattern is typical: in the vast majority of cases, the justification for the bootstrap comes from the validity of some asymptotic approximation to the test statistic, so the asymptotic method could have been used as well. There are only a couple of known cases in which we do not know whether a statistic has an asymptotic limit and, if so, what it is, but the bootstrap magically works. And recall from the discussion of when the bootstrap fails in Lecture Note 4 that in the well-known list of settings where there are difficulties in carrying out the asymptotic approach, the bootstrap does not work either.

Fourth, another thing the bootstrap is implicitly doing here is bias correction. Recall from Lecture Note 5 that a plug-in estimator such as $\hat{\gamma} = h(\bar{X}_n, s^2)$ typically carries a bias of order $O(1/n)$ arising from the nonlinearity of h , and that the bootstrap distribution reproduces this bias. The re-centered bootstrap distribution of z^* therefore reflects the $O(1/n)$ bias present in the sampling distribution of z , which the normal approximation $N(0, (\nabla h)' V \nabla h)$ ignores.

Second-order refinements. In the examples above, by using the bootstrap we avoided calculating a standard error for $\hat{\gamma}$. The question arises: if we are willing to calculate standard errors,

would using the bootstrap still have any benefits? The answer is yes. Imagine we construct a natural consistent estimator $\hat{V}$ for V and consider the t -statistic

$$t = \sqrt{n} \frac{\hat{\gamma} - 0}{\sqrt{(\nabla \hat{h})' \hat{V} \nabla \hat{h}}},$$

where $\nabla \hat{h}$ is the corresponding derivative evaluated at $(\bar{X}_n, s^2)$. We know that under the null $t \xrightarrow{d} N(0, 1)$. The bootstrapped statistic t^* is constructed in an analogous way (using re-centering), and $t^* \xrightarrow{d} N(0, 1)$ as well. Thus the distribution of t is close to the distribution of t^* in large samples. However, there are results suggesting that the distance between the distributions of t and t^* is asymptotically smaller than the distance between the distribution of t and the standard normal cdf. That is, the bootstrap provides a better approximation, a so-called *second-order refinement*.

Statements like this one are based on the *Edgeworth expansion*, which in general is quite complicated to derive and which we will not derive here. It is a statement of the following sort:

$$P\{t \leq x\} = \Phi(x) + \frac{1}{\sqrt{n}} \phi(x, F) + O\left(\frac{1}{n}\right),$$

where the correction term $\phi(x, F)$ depends on the underlying distribution F only through a few of its lowest moments (in our example, roughly the first six). Similarly, under the bootstrapped distribution,

$$P^*\{t^* \leq x\} = \Phi(x) + \frac{1}{\sqrt{n}} \phi(x, \hat{F}_n) + O_p\left(\frac{1}{n}\right).$$

One may show that $\phi(x, \hat{F}_n) - \phi(x, F) = O_p(1/\sqrt{n})$ (by the CLT and the delta method, since the moments of $\hat{F}_n$ are within $O_p(1/\sqrt{n})$ of the corresponding moments of F). Then

$$P\{t \leq x\} - P^*\{t^* \leq x\} = O_p\left(\frac{1}{n}\right),$$

while

$$P\{t \leq x\} - \Phi(x) = O\left(\frac{1}{\sqrt{n}}\right).$$

Note that the refinement is obtained by bootstrapping the asymptotically pivotal statistic t , whose limit distribution $N(0, 1)$ is free of unknown parameters — one more argument in favor of basing tests on (asymptotically) pivotal statistics.